

Agentic Payments, Billing & Consumer Dispute Resolution Platform

Use case document and build starter resources

Use case theme	Improving the buying experience with agentic AI in customer service
Primary domain	Customer service, financial operations, billing, refunds, chargebacks, consumer dispute resolution, and fraudulent user request detection
Version	1
Date	June 8, 2026
Executive positioning This platform turns customer support from reactive ticket handling into an intelligent financial-resolution layer. It identifies billing and transaction issues, detects potentially fraudulent user requests before sensitive actions, gathers evidence, applies policy and regulatory constraints, proposes remedies, executes approved actions, and keeps an auditable case history across payments, billing, CRM, risk, and customer channels.	

Note: Regulatory references are included for planning purposes and must be validated by legal/compliance teams for each jurisdiction before production launch.

Contents

- 1. Executive summary
- 2. Use case review and refined concept
- 3. Market need and opportunity
- 4. Target users and service scenarios
- 5. Proposed solution capabilities, including fraudulent-request detection
- 6. Examples of customer journeys
- 7. Recommended agent topology
- 8. Data model, integrations, and tool catalog
- 9. Decision boundaries, fraud controls, and compliance
- 11. Evaluation
- 12. Available resources to start building
- Appendix A. Hackathon scope and judging rubric

1. Executive summary

The proposed platform is an intelligent financial operations agent for customer service. It handles payments, billing issues, subscription changes, refunds, chargebacks, consumer disputes, and suspicious user requests across e-commerce, telecom, utilities, subscriptions, marketplaces, banks, fintechs, and payment providers. The same workflow that accelerates legitimate remedies should also detect refund abuse, account-takeover indicators, duplicate-refund attempts, false item-not-received claims, and other fraudulent requests before irreversible customer, account, or payment actions occur.

Core value hypothesis

Customers get faster answers and remedies. Support teams spend less time on repetitive investigation and handoff work. Finance, dispute, and fraud teams receive cleaner evidence packages, stronger audit trails, and earlier risk signals before a suspicious request becomes refund leakage, account harm, or a formal chargeback/regulatory escalation.

Dimension	Recommended position
Product category	Agentic customer-service and financial-operations platform
Primary jobs to be done	Understand the issue, verify the customer, assess request legitimacy and fraud risk, gather payment/order/billing facts, apply policies, recommend or execute remedies, document the case, and monitor closure.
Best initial market	High-volume digital commerce or subscription businesses with frequent refund, late delivery, duplicate charge, failed payment, cancellation, and billing adjustment tickets.
Initial wedge	Read-only investigation plus human-approved refunds/credits for e-commerce and subscription billing; later expand to autonomous low-risk actions and dispute evidence submission.
Differentiator	Autonomous multi-step execution across systems, not just chat responses: the agent can reason over policy, retrieve evidence, detect risk patterns, call approved tools, and maintain an auditable financial case file.
Risk posture	Human-in-the-loop by default for regulated, high-value, ambiguous, fraud-related, suspected account-takeover, or customer-harm decisions; strict tool permissions and audit logs for every action.

The use case is well suited to agentic AI because dispute resolution is rarely a single question-answer turn. It requires multi-step reasoning, data retrieval, evidence gathering, policy interpretation, customer communication, system updates, and follow-up monitoring. Traditional chatbots can deflect simple questions, but they struggle when the resolution requires coordinated action across multiple systems.

2. Use case review and refined concept

2.1 Use case statement

When a customer raises a payment, billing, subscription, refund, chargeback, or dispute issue, an AI agent verifies identity and consent, retrieves relevant account and transaction facts, screens for potentially fraudulent or abusive request patterns, interprets approved policies and timelines, proposes eligible remedies, executes low-risk actions or routes exceptions for human approval, and provides the customer with a clear resolution and follow-up path.

2.2 What makes this agentic

- **Goal-directed:** the system pursues an end-to-end resolution objective, not just a response.
- **Tool-using:** it securely calls APIs for CRM, orders, billing, subscriptions, payment processors, identity verification, messaging, and case management.
- **Stateful:** it maintains workflow state, deadlines, evidence, approvals, and customer communication history.
- **Policy-grounded:** responses and actions are constrained by current company policy, contractual terms, processor/network rules, and regulatory timelines.
- **Auditable:** every decision, data source, tool call, approval, and customer notice is recorded for compliance and operational review.
- **Risk-aware:** the agent evaluates identity confidence, account history, velocity, behavioral anomalies, evidence consistency, and duplicate-refund or double-dip patterns before recommending customer-impacting actions.

3. Market need and opportunity

The market context is favorable: customer-service automation is moving from narrow FAQ chatbots toward agents that can complete multi-step tasks. Gartner predicts that by 2029, agentic AI will autonomously resolve 80% of common customer service issues and reduce operational costs by 30% [1]. Earlier chatbot experiences created a trust gap, with UJET reporting in 2022 that 80% of consumers who interacted with a chatbot said it increased frustration [3].

The payments and dispute domain adds urgency. Mastercard's 2025 chargeback outlook, based on Datos Insights research, forecasts global chargeback volume growing from 261 million transactions in 2025 to 324 million in 2028, while noting that traditional dispute management processes struggle with growing volume and that automated and AI-based tools can reduce friction and improve the customer experience [4].

Pain point	Business impact	Agentic AI opportunity
Fragmented systems	Support agents jump across CRM, OMS, billing, PSP dashboards, shipment tracking, and policy docs.	Unify retrieval and action execution through governed tools.
Slow resolution	Customers wait while teams investigate, transfer, and ask for repeated information.	Auto-gather facts and present eligible remedies in the same conversation.
Chargeback escalation	Unresolved merchant support issues can turn into card disputes, fees, lost revenue, and evidence workload.	Resolve merchant-level complaints earlier and package evidence when a formal dispute occurs.
Policy inconsistency	Different agents may offer different remedies for similar facts.	Use a versioned policy engine and structured decision output.
Compliance risk	Regulatory deadlines and customer notices can be missed without strong workflow management.	Use SLA timers, required disclosures, approval gates, and audit logs.
Fraudulent or abusive requests	Refund abuse, false item-not-received claims, account takeover, and double-dip refund/chargeback attempts can create direct loss and customer harm.	Score request risk, block or pause suspicious actions, require step-up verification, and route cases to fraud/risk review.

3.1 Strategic opportunity

- **Customer trust:** provide clear explanations, transparent evidence, and timely remedies.
- **Operational efficiency:** reduce repetitive investigation and manual case preparation.
- **Revenue protection:** prevent avoidable refunds/chargebacks while honoring legitimate customer remedies.
- **Fraud prevention:** detect suspicious refund, credit, cancellation, and dispute requests without making unsupported accusations to the customer.
- **Compliance readiness:** capture immutable case records, approvals, decision logic, and customer notices.
- **Data flywheel:** learn from dispute reasons, policy gaps, merchant descriptor confusion, subscription cancellation friction, and fulfillment failures.

4. Target users and service scenarios

Persona	Need	Agent role	Success outcome
Consumer / buyer	Resolve a charge, refund, subscription, late shipment, or overbilling issue without repeating information.	Conversational guide, investigator, and resolution coordinator.	Clear remedy, timeline, confirmation, and follow-up.
Customer service agent	Handle escalations and exceptions with context and recommended next action.	Assistant that summarizes facts, drafts responses, and prepares tool actions.	Lower handle time and more consistent outcomes.
Finance / billing ops	Apply credits, refunds, reversals, write-offs, and invoice corrections accurately.	Prepares calculations, validates eligibility, and logs approved actions.	Cleaner audit trail and fewer manual errors.
Dispute analyst	Prepare chargeback evidence and decide whether to accept, refund, or represent.	Evidence collector and representment package drafter.	Faster case preparation and better case selection.
Risk/compliance	Ensure policy, regulatory, and customer-harm controls are met.	Policy gatekeeper and audit event generator.	Defensible, traceable decisions.
Risk / fraud analyst	Detect abuse patterns, account takeover, serial refund requests, false claims, and duplicate recovery attempts.	Risk investigator that reviews flagged cases, evidence, and recommended controls.	Lower fraud leakage and fewer unfair customer decisions.

4.1 Priority service scenarios

Scenario	Customer trigger	Agent outcome	Initial fit
Late e-commerce order	Package has not arrived by promised date.	Checks order/tracking, confirms delay, offers reshipment, refund, or discount per policy.	High
Duplicate or unrecognized charge	Customer sees two charges or does not recognize descriptor.	Matches payment to order, identifies duplicate/auth hold/descriptor confusion, issues refund or explanation.	High
Subscription cancellation / renewal dispute	Customer claims they cancelled or did not expect renewal.	Checks cancellation history, renewal notices, terms, and refund eligibility; cancels or credits if eligible.	High
Telecom/utility overcharge	Customer claims plan, usage, tax, fee, or meter/billing error.	Compares bill to plan/rate tables and creates adjustment case with evidence.	Medium
Formal card dispute / chargeback	Issuer/processor notifies merchant of dispute.	Classifies reason code, gathers evidence, drafts response, routes to human approval.	Medium
Suspicious refund or dispute request	Customer requests a refund/credit while signals show unusual velocity, account changes, inconsistent evidence, or prior recovery for the same transaction.	Pauses automated refund, performs risk checks, requests step-up verification if appropriate, and routes to fraud/risk review with a structured evidence summary.	High

5. Proposed solution capabilities, including fraudulent-request detection

The platform should be designed as a layered agentic system. The customer-facing experience can remain conversational, but financial decisions and actions must be deterministic, policy-constrained, logged, and reversible where possible.

Capability	What it does	Build notes
Omnichannel intake	Accepts requests from chat, voice, email, SMS/WhatsApp, mobile app, web portal, and agent desktop.	Start with authenticated web/mobile chat or support portal to reduce identity risk.
Identity, consent, and authorization	Verifies customer, account ownership, channel risk, and permission to act.	Use step-up verification for refunds, credential changes, high-value cases, or suspicious behavior.
Case classification	Classifies issue type, urgency, risk, regulatory category, and required evidence.	Use structured outputs; never rely on free-form text alone for routing.
Evidence retrieval	Collects order, invoice, payment, subscription, shipment, usage, prior contact, and policy evidence.	Apply data minimization; mask PII and tokenized payment identifiers.
Policy and eligibility engine	Determines allowed remedies and required approvals based on policies, contracts, and timelines.	Keep rules versioned and separate from model prompts.
Resolution recommendation	Generates customer-friendly options and internal rationale.	Ground recommendations in retrieved facts and cite policy snippets internally.
Tool execution	Calls approved APIs for refunds, credits, order replacement, subscription changes, ticket updates, and notifications.	Use allowlisted tools, idempotency keys, transaction limits, and approval gates.
Human-in-the-loop	Routes exceptions and sensitive actions for human review.	Required for high-value refunds, chargeback submissions, regulated complaints, fraud, legal threats, and policy exceptions.
Audit and monitoring	Records actions, model outputs, sources, decisions, approvals, and customer notices.	Use immutable event log and exportable case file for compliance.
Fraudulent user request detection	Detects suspicious refund, credit, cancellation, account-change, and dispute requests using identity confidence, transaction facts, case history, velocity, device/channel signals, and evidence consistency.	Do not let the LLM make final fraud determinations alone; use deterministic risk rules, approved risk models, step-up verification, and human review for adverse decisions.

5.1 Fraudulent user request detection and response

Fraud detection should be a first-class part of the use case, not an afterthought. The agent should classify suspicious requests as risk signals that need verification or review, while avoiding unsupported accusations in customer-facing language. The objective is to protect customers and the business while still resolving legitimate issues quickly.

Fraud pattern	Indicative signals	Agent response
Refund abuse / serial claims	High refund or credit velocity, repeated similar claims, mismatched item/order details, unusual goodwill requests.	Pause automated refund; show reason codes internally; route to risk review when threshold is exceeded.
Account takeover or social engineering	New device/channel, failed or unusual authentication, recent account/email/address changes, pressure for urgent monetary action.	Require step-up verification; block sensitive changes until identity confidence is sufficient.
False item-not-received or service-not-delivered claim	Delivery proof, usage logs, activation records, or prior communications conflict with the claim.	Prepare evidence summary; offer policy-compliant appeal path; require human review before denial.
Double-dip recovery	Refund already issued, credit already applied, or active chargeback exists for the same transaction.	Block duplicate refund/credit; escalate to dispute specialist; prepare evidence showing prior remedy.
Manipulated or inconsistent evidence	Screenshots, attachments, order IDs, timestamps, or narratives conflict with system-of-record data.	Treat attachments as untrusted; request additional verification or route to manual evidence review.
Bot or scripted abuse	High-volume repeated requests, similar language across accounts, disposable emails, abnormal channel velocity.	Rate-limit, require stronger authentication, and create a fraud investigation case.

Design principle: the customer-facing agent should say that the request needs additional verification or specialist review, while the internal case file records the risk rationale, evidence, score, and reviewer decision.

6. Examples of customer journeys

6.1 Example journey: late e-commerce order

- **Customer:** “My order is late and I need it before Friday.”
- **Agent:** Verifies identity, pulls order and shipment data, sees carrier delay and delivery risk.
- **Agent:** Checks fulfillment and refund policy. Determines customer is eligible for free reshipment, full refund after return, or courtesy discount.
- **Customer:** Chooses replacement shipment.
- **Agent:** Creates replacement order, adds expedited shipping if allowed, sends tracking confirmation, and schedules a check-in.
- **System:** Logs evidence, decision rule, customer selection, order action, confirmation message, and follow-up reminder.

6.2 Example journey: duplicate charge

The agent identifies the relevant payment intent/charge, compares authorization and capture states, detects whether the duplicate is a pending authorization, settled duplicate, split shipment, or legitimate recurring payment, and either explains the expected authorization release, issues a refund, or escalates to a billing specialist. The output must include a transaction-level audit trail and a customer-facing explanation that avoids exposing sensitive payment data.

6.3 Example journey: subscription renewal dispute

The agent checks renewal notice history, cancellation events, plan terms, last login/activity signals, prior complaints, payment status, and refund policy. It then proposes cancellation, prorated credit, one-time courtesy refund, or escalation when the case touches legal, regulatory, or high-value thresholds.

6.4 Example journey: suspicious refund request

- **Customer:** Requests an immediate refund for a high-value order and says the item was not received.
- **Agent:** Verifies identity and retrieves order, payment, shipment, refund, chargeback, and prior case history.
- **Agent:** Detects risk signals: delivery confirmation exists, the account changed its shipping address recently, and a previous goodwill credit was issued for the same order.
- **Agent:** Pauses autonomous refund execution, requests step-up verification if appropriate, and creates an internal risk summary with supporting evidence.
- **Human risk/review specialist:** Approves a remedy, requests more evidence, denies the request with an approved appeal path, or escalates to dispute/compliance workflows.
- **System:** Logs the risk score, signals, evidence sources, verification outcome, reviewer decision, customer notice, and any follow-up actions.

7. Recommended agent topology

Agent / service	Responsibility	Example tools
Triage agent	Classify issue, risk, jurisdiction, customer intent, and urgency.	classify_case, detect_regulatory_trigger, detect_fraud_signal
Payments agent	Retrieve payment details, identify authorization/capture/refund status, and draft payment actions.	get_payment, create_refund_request, check_refund_status
Billing/subscription agent	Investigate invoices, plans, usage, renewals, cancellations, and credits.	get_invoice, get_subscription, calculate_adjustment, cancel_subscription
Fulfillment agent	Check order status, shipping events, return status, and	get_order, get_tracking,

	replacement availability.	create_replacement_order
Dispute evidence agent	Assemble evidence packages for processor/card/PayPal disputes and summarize rationale.	get_dispute, gather_evidence, draft_representment
Compliance guardrail service	Validate action eligibility, thresholds, customer communications, and audit requirements.	validate_policy, require_approval, check_sla, log_audit
Fraud/risk agent or service	Evaluate suspicious user requests, account-takeover indicators, refund abuse, duplicate recovery attempts, and evidence anomalies.	get_risk_profile, evaluate_request_risk, check_velocity, create_fraud_review

8. Data model, integrations, and tool catalog

8.1 Core data entities

Entity	Purpose	Important fields
Customer	Identifies the person or business and communication preferences.	customer_id, verified_identity_level, contact_channels, consent_status
Account / subscription	Defines active services, plans, entitlements, cancellation state, and billing cycle.	account_id, plan_id, renewal_date, cancellation_events, usage_summary
Transaction / payment	Represents payment events across PSPs and ledgers.	payment_id, amount, currency, status, processor, charge_id, payment_intent_id, descriptor
Order / shipment	Links purchase, fulfillment, delivery, return, and replacement state.	order_id, tracking_id, promised_date, carrier_status, return_status
Case / dispute	Tracks complaint, customer intent, issue category, workflow state, SLA, and resolution.	case_id, reason_code, jurisdiction, risk_score, SLA_due_at, status
Evidence item	Records source documents and facts used to support decisions.	evidence_id, type, source_system, timestamp, hash, redaction_status
Decision record	Captures agent recommendation, policy rationale, approval path, and action result.	decision_id, policy_version, confidence, approver_id, execution_status
Audit event	Immutable timeline of system, agent, customer, and human actions.	event_id, actor, action, source, tool_call_id, created_at
Risk signal / fraud case	Captures suspicious request indicators and review outcomes without exposing unnecessary sensitive data.	risk_signal_id, signal_type, risk_score, velocity_window, reviewer_outcome, appeal_status

8.2 Integration map

Integration	Purpose	Initial priority	Build resource
CRM / support desk	Tickets, comments, customer profile, escalations, macros.	High	Zendesk Tickets/Webhooks [16] or existing CRM APIs
Payment service provider	Payment lookup, refunds, disputes, webhooks, evidence submission.	High	Stripe [11][12][13], Adyen [14], PayPal [15]
Order management / fulfillment	Order status, delivery dates, return/refund state, reshipment.	High	Internal OMS, Shopify/commerce APIs, carrier tracking APIs
Identity verification	Step-up auth, account ownership, fraud reduction.	High	Twilio Verify [18], Plaid IDV [20], existing IAM/MFA
Billing / subscription	Invoices, credits, renewals, plan usage, cancellation events.	High	Stripe Billing [13] or internal billing/ERP APIs
Knowledge/policy repository	Grounded answers and deterministic policy decisions.	High	Versioned policy docs, RAG index, policy-rule engine
Dispute processors/network tools	Formal chargeback events, representment, alert/prevention workflows.	Medium	Adyen/Stripe/PayPal APIs; Verifi/Ethoca depending on contracts

Observability / audit	Trace agent runs, tool calls, approvals, and outcomes.	High	OpenAI tracing [10], SIEM/log platform, data warehouse
Fraud/risk systems	Risk score, velocity checks, account-takeover signals, device/session intelligence, fraud case creation.	High	Internal fraud engine, PSP risk tools, identity/MFA logs, device/session telemetry.

8.3 Tool catalog and authority levels

Tool	Purpose	Read/write	Approval	Notes
get_customer_profile	Retrieve profile, entitlements, history.	Read	No	Mask sensitive fields by default.
verify_identity	Start or check step-up verification.	Write	Conditional	Required for account or money movement.
get_payment_details	Retrieve payment status, amount, dates, processor IDs.	Read	No	Never expose raw PAN.
get_order_status	Retrieve order, shipment, return, delivery promise.	Read	No	Include source timestamp.
calculate_refund_eligibility	Apply policy to determine allowed remedies.	Read	No	Deterministic policy service, not free-form model logic.
create_refund	Create full or partial refund.	Write	Yes above limit	Use idempotency key, amount cap, and policy version.
issue_account_credit	Apply courtesy or billing credit.	Write	Yes above limit	Separate accounting code and reason.
cancel_subscription	Cancel or schedule cancellation.	Write	Conditional	Require clear customer confirmation.
submit_dispute_evidence	Submit chargeback/claim evidence.	Write	Yes	Human approval required initially.
send_customer_message	Send resolution or follow-up notification.	Write	No/conditional	Template-controlled for regulated notices.
escalate_case	Assign to human agent/specialist.	Write	No	Mandatory for risk triggers.
log_audit_event	Create immutable event record.	Write	No	Called for every step.
get_risk_profile	Retrieve risk history, known flags, prior refunds/disputes, velocity, and verification status.	Read	No	Mask sensitive fields; use only for authorized cases.
evaluate_request_risk	Score the current request for refund abuse, ATO, duplicate recovery, or evidence inconsistency.	Write	Conditional	Creates auditable risk assessment; high-risk scores force escalation.
create_fraud_review	Open a specialist review case with evidence, risk signals, and recommended controls.	Write	No	Required before adverse action when risk threshold is exceeded.

9. Decision boundaries, fraud controls, and compliance

The platform should be launched with explicit boundaries. The agent should be allowed to explain, retrieve, summarize, recommend, and prepare actions broadly, but direct financial execution should begin with low-risk actions and be expanded only after evaluation and governance milestones are met.

Autonomy level	Allowed examples	Required controls
Level 0 - Inform	Order/payment status, policy explanation, invoice copy, next steps.	Authenticated account or safe public policy only; no action.
Level 1 - Prepare	Draft refund request, evidence packet, customer message, case summary.	Grounding sources, human review for external/customer-impacting output.
Level 2 - Execute low risk	Resend receipt, update ticket, schedule follow-up, apply small courtesy credit, cancel per explicit request.	Policy check, confirmation, idempotency, audit log, low amount threshold, and no active fraud/risk flag.

Level 3 - Execute medium risk	Partial refund, replacement order, invoice adjustment, subscription proration.	Step-up auth, amount threshold, policy validation, risk score check, no unresolved fraud flag, and possible human approval.
Level 4 - Human-approved regulated/high risk	Chargeback evidence submission, final denial, suspected fraudulent request, fraud allegation, legal/regulatory complaint, high-value refund.	Mandatory human approval, risk/compliance template, record retention, customer appeal path, and escalation SLA.

9.1 Compliance considerations

Area	Why it matters	Planning resource
Consumer electronic fund transfer errors	Debit/EFT error-resolution workflows can carry specific notice, investigation, provisional credit, and timing obligations.	CFPB Regulation E procedures [5].
Credit-card billing disputes	Credit disputes and billing errors have written notice and timing requirements under Regulation Z/FCBA.	CFPB Regulation Z [6] and FTC FCBA summary [7].
Payment data security	Avoid raw card data, tokenize payment identifiers, and scope PCI obligations carefully.	PCI Security Standards Council resources [25].
AI governance	Agentic systems need risk management, transparency, monitoring, and accountability controls.	NIST AI RMF [21], NIST GenAI Profile [22], ISO/IEC 42001 [26].
LLM/application security	Prompt injection, sensitive data leakage, excessive agency, and tool misuse are high-priority risks.	OWASP LLM/Agentic resources [23] [24].
Privacy and retention	Customer, transaction, dispute, and communication data may contain PII and financial information.	Data minimization, retention schedule, encryption, access control, DPA review.
Fraud, abuse, and adverse action governance	Suspicious user requests require careful handling to avoid unfair denial, privacy issues, unsupported accusations, or inconsistent treatment.	Risk policy, adverse-action/appeal playbook, model monitoring, and human review requirements.

9.2 Key risk controls

- **Prompt-injection defense:** treat customer text, documents, emails, and web content as untrusted; never allow retrieved text to change tool permissions or policy.
- **Least-privilege tools:** give the agent only scoped APIs and use separate read, prepare, and execute tools.
- **Deterministic financial checks:** validate amount, currency, account, policy version, and idempotency before any refund/credit.
- **Human approvals:** require approval for high-risk actions, unclear policy, high-value refunds, formal disputes, legal threats, vulnerable customers, suspected fraudulent user requests, fraud allegations, or repeated complaints.
- **Immutable audit:** store case facts, retrieved evidence, prompts/outputs where permitted, tool calls, approvals, and customer notifications.
- **Step-up verification:** require MFA or identity verification for sensitive account changes and monetary actions.
- **Model evaluations:** run regression tests against historical cases, synthetic edge cases, fraudulent request patterns, prompt injection, policy changes, and adversarial customer behavior.

9.3 Fraudulent user request detection controls

Control	Purpose	Operating rule
Risk scoring before money movement	Every refund, credit, cancellation reversal, or dispute remedy request should be checked for request-level risk before execution.	Block or require approval when score exceeds threshold.
Step-up identity verification	Suspicious requests should require MFA, identity verification, or secure account re-authentication before sensitive data is exposed or changed.	Escalate failed or inconsistent verification.
Velocity and pattern detection	Track refund counts, dispute frequency, account changes, repeated language, shared identifiers, and channel velocity.	Use cooldowns, rate limits, and review queues.
Duplicate recovery prevention	Detect when the same transaction already has a refund, credit, adjustment, chargeback, or open	Do not issue duplicate remedies; prepare evidence instead.

	dispute.	
Fairness and customer protection	Risk flags should not become automatic accusations or denials without evidence and review.	Use approved wording, explain next steps, and provide an appeal path.
Audit and feedback loop	Store signals, score, evidence, reviewer outcome, and downstream loss/recovery result.	Use outcomes to tune rules, thresholds, and evaluation tests.

10. Evaluation

10.1 Why AI agent evaluation is different

Traditional AI evaluation assumes a comparatively simple input-output setting: the model receives a prompt, produces an answer, and the evaluator scores that answer. Agentic systems are different. They operate in closed loops, maintain state, invoke tools, interact with external services, coordinate with other agents, and may continue acting after the first response. Farooq et al. define agentic AI systems as LLM-based systems that maintain state, plan over multiple steps, use tools, and adapt dynamically in open-ended environments [30]. Baduwal and Paudel similarly describe agentic AI as multi-step, tool-using, adaptive, long-horizon decision-making systems [35].

This creates several evaluation challenges. First, final success can hide unsafe or unreliable behavior. Gritta et al. show that agents can skip critical steps, simulate or hallucinate tool use, rely on outdated memory, and still produce plausible outputs [34]. Second, agent execution is non-deterministic. Moshkovich et al. report substantial variability in execution flow, cost, latency, and LLM calls even for repeated runs of the same task, motivating behavioral benchmarking and trace analytics [31]. Third, many failures occur at component boundaries: tool selection, parameter mapping, memory retrieval, environment authorization, and inter-agent coordination [33].

The practical implication is that agent evaluation must ask two questions at the same time: did the agent accomplish the user goal, and did it accomplish the goal through a reliable, safe, cost-effective, and auditable process? For this dispute-resolution platform, both questions are required for deployment readiness.

10.2 Evaluation principles for the agentic platform

This platform's dispute-resolution agent should be evaluated as an interactive system that perceives, reasons, plans, uses tools, maintains memory, acts in an environment, and adapts over time. The agent-evaluation literature consistently argues that ordinary single-turn model metrics such as exact-match accuracy, BLEU/ROUGE, or final task success are necessary but insufficient. They miss whether an agent followed the correct procedure, chose the right tools, used memory appropriately, recovered from errors, obeyed policies, and produced auditable evidence of its actions [30], [33], [34], [35].

The strongest evaluation practice is therefore portfolio-based and lifecycle-aware. A credible evaluation combines public benchmarks, private domain tasks, component tests, process checklists, trace observability, safety and adversarial tests, calibrated human or AI judges, repeated stochastic runs, and post-deployment monitoring. This is especially important because agents can succeed by shortcuts, fail non-deterministically, hallucinate tool use, rely on stale parametric knowledge, or violate policies while still appearing useful [31], [32], [34].

Key recommendations

- **Evaluate trajectories, not only outputs.** Capture the full sequence of prompts, plans, tool calls, observations, memory reads/writes, retries, errors, approvals, and final outputs. Use this trace to compute component and process metrics [31], [33], [34].
- **Use a multidimensional scorecard.** Report task success, partial progress, output quality, planning quality, tool reliability, memory retrieval, process compliance, safety, robustness, autonomy, efficiency, collaboration, observability, and user utility separately [30], [35].
- **Add process compliance.** For high-risk or regulated workflows, define a recommended protocol and score whether the agent followed it using unweighted and weighted compliance scores [34].
- **Audit benchmark validity.** Before reporting leaderboard-style scores, classify benchmarks by contamination risk, temporal holdout, environmental isolation, and vulnerability to gaming or shortcut learning [30].
- **Calibrate judges.** LLM-as-Judge and Agent-as-a-Judge can scale evaluation, but should be checked against human consensus, especially for domain-specific or high-stakes use cases [32], [34].
- **Instrument for production.** Use observability conventions, trace logs, audit artifacts, and runtime SLOs to diagnose failures that black-box benchmarking cannot reveal [31], [33].

10.3 Evaluation plan

The evaluation plan operationalizes the principles in Section 10.2. Each activity is aligned to one or more of the key recommendations and preserves the platform's domain-specific testing for disputes, refunds, and fraud.

- **Trajectory capture and offline replay (evaluate trajectories, not only outputs):** replay historical support tickets and compare agent recommendations to actual human outcomes and approved policy, capturing the full trace of prompts, plans, tool calls, observations, memory reads/writes, retries, errors, approvals, and final outputs so component and process metrics can be computed rather than final resolution accuracy alone [31], [33], [34].
- **Multidimensional scoring against the KPI scorecard (use a multidimensional scorecard):** score every replayed and synthetic case across the dimensions in the metric table above — task success, partial progress, output quality, planning quality, tool reliability, memory retrieval, process compliance, safety, robustness, autonomy, efficiency, collaboration, observability, and customer utility — and report them separately instead of collapsing to a single pass/fail [30], [35].
- **Process-compliance and tool-call validation (add process compliance):** for high-risk or regulated workflows such as Reg E/Z disputes, chargebacks, and high-value refunds, define a recommended protocol and score adherence with unweighted and weighted compliance scores; validate every tool for schemas, idempotency, amount caps, permissions, and failure handling; and generate synthetic edge cases for ambiguous policy, partial refunds, duplicate charges, chargeback reason codes, vulnerable customers, and legal threats [33], [34].
- **Dataset and benchmark validity audit (audit benchmark validity):** before reporting any aggregate or leaderboard-style scores, classify the redacted-ticket and synthetic datasets by contamination risk, temporal holdout, environmental isolation, and vulnerability to gaming or shortcut learning so reported numbers reflect real capability rather than leaked or memorized cases [30].
- **Judge calibration and human-review quality (calibrate judges):** where LLM-as-Judge or Agent-as-a-Judge grades recommendations at scale, validate it against human reviewer consensus on a sampled set, especially for domain-specific or high-stakes cases, and track how often reviewers accept, edit, reject, or escalate agent recommendations as the human-consensus signal [32], [34].
- **Production instrumentation, safety, and adversarial monitoring (instrument for production):** apply observability conventions, trace logs, audit artifacts, and runtime SLOs to diagnose failures black-box benchmarking cannot reveal; red-team prompt injection through customer messages, attachments, policy documents, email threads, and merchant descriptors; run fraud/abuse testing for serial refund attempts, double-dip refund plus chargeback, account takeover, fake item-not-received claims, manipulated evidence, and coordinated bot-like requests; and stage rollout as shadow mode, then human-approved execution, then limited autonomous low-risk actions after governance approval [31], [33].

11. Available resources to start building

The following resources are available or readily obtainable to begin design, prototyping, and pilot implementation. The team should start with sandbox integrations, synthetic cases, and redacted historical tickets before connecting to live financial systems.

Resource area	Recommended resource	How to use it	Source / note
AI agent orchestration	OpenAI Agents SDK / Responses API or LangGraph for graph-style orchestration; CrewAI for role-based multi-agent crews; Microsoft Agent Framework (open-source successor to AutoGen and Semantic Kernel); Google Agent Development Kit (ADK); and Anthropic's Claude Agent SDK.	Build triage and specialist agents, tool calls, handoffs, guardrails, HITL, and traces; evaluate several open-source orchestrators against the workflow before standardizing on one.	OpenAI [8][9][10], LangGraph [27], CrewAI [36], Microsoft Agent Framework [37], Google ADK [38], Claude Agent SDK [39].
Durable workflows	Temporal or equivalent workflow engine.	Persist long-running dispute workflows, SLA timers, retries, and approval pauses.	Temporal Python docs [28].
Knowledge and policy grounding	Versioned policy repository plus retrieval index.	Store refund policies, terms, billing playbooks, templates, regulatory workflows, and evidence checklists.	Use pgvector/Postgres or existing search stack [29].
Payment/refund APIs	Stripe sandboxes, Adyen test account, PayPal sandbox.	Look up transactions, create refunds in sandbox, receive dispute webhooks, and draft evidence responses.	Stripe [11][12][13], Adyen [14], PayPal [15].
CRM/contact center	Zendesk, Salesforce Service Cloud, Twilio Flex,	Create/update cases, route escalations,	Zendesk [16], Twilio [17].

	or existing service desk.	preserve conversation context, and surface approvals to agents.	
Identity verification	Existing IAM/MFA, Twilio Verify, Plaid Identity Verification.	Step-up verification before account or monetary actions.	Twilio [18], Plaid [20].
Financial/account data	Plaid Transactions where user-authorized account data is needed, internal ledger, billing, ERP.	Investigate transaction history, merchant identity, amounts, and related billing records.	Plaid [19].
Regulatory and governance	Reg E/Z workflows, PCI resources, NIST AI RMF, OWASP LLM/Agentic security, ISO 42001.	Define control requirements, threat model, risk acceptance, and audit expectations.	CFPB/FTC [5][6][7], PCI [25], NIST [21][22], OWASP [23] [24], ISO [26].
Evaluation data	Redacted tickets, synthetic disputes, policy examples, sandbox transactions, test personas.	Build regression test suite and benchmark agent recommendations.	Internal data plus synthetic generation; privacy review required.
Operational templates	Escalation matrix, approval thresholds, refund reason codes, message templates, audit checklist.	Constrain the agent and reviewers with consistent, approved workflows.	Create during discovery phase.
Fraud/risk signal package	Historical fraud/refund-abuse labels, velocity counters, account-change logs, device/session signals, PSP risk outputs, chargeback/refund history.	Create risk features, threshold rules, escalation reasons, and false-positive review reports for the pilot.	Internal risk/fraud data, PSP risk tools, identity/MFA logs, and approved fraud taxonomy.

References

- [1] [Gartner press release: Agentic AI will autonomously resolve 80% of common customer service issues by 2029](#)
- [2] [Forbes: IBM Watson and Go Moment Ivy in hospitality customer service](#)
- [3] [BusinessWire / UJET: 80% of consumers report increased frustration with chatbots](#)
- [4] [Mastercard: 2025 State of Chargebacks Report](#)
- [5] [CFPB Regulation E, Section 1005.11: Procedures for resolving errors](#)
- [6] [CFPB Regulation Z, Section 1026.13: Billing error resolution](#)
- [7] [FTC: Fair Credit Billing Act summary](#)
- [8] [OpenAI Agents SDK guide](#)
- [9] [OpenAI Agents SDK: Human-in-the-loop](#)
- [10] [OpenAI Agents SDK: Tracing](#)
- [11] [Stripe API Reference: Disputes](#)
- [12] [Stripe API Reference: Refunds](#)
- [13] [Stripe Billing APIs](#)
- [14] [Adyen Docs: Dispute webhooks](#)
- [15] [PayPal Developer: Customer Disputes API](#)
- [16] [Zendesk Developer Docs: Tickets API](#)
- [17] [Twilio Flex documentation](#)
- [18] [Twilio Verify API](#)
- [19] [Plaid Transactions API](#)
- [20] [Plaid Identity Verification API](#)
- [21] [NIST AI Risk Management Framework](#)
- [22] [NIST Artificial Intelligence Risk Management Framework: Generative AI Profile](#)
- [23] [OWASP LLM06:2025 Excessive Agency](#)
- [24] [OWASP Agentic Security Initiative](#)
- [25] [PCI Security Standards Council](#)
- [26] [ISO/IEC 42001:2023 AI management systems](#)
- [27] [LangGraph persistence documentation](#)
- [28] [Temporal Python SDK workflows documentation](#)

[29] [pgvector: vector similarity search for PostgreSQL](#)

[30] A. Farooq, S. Raza, N. Karim, H. Iqbal, A. V. Vasilakos, and C. Emmanouilidis, "Evaluating and regulating agentic AI: A study of benchmarks, metrics, and regulation," *Information Fusion*, vol. 136, Article 104444, 2026. doi: 10.1016/j.inffus.2026.104444.

[31] D. Moshkovich, H. Mulian, S. Zeltyn, N. Eder, I. Skarbovsky, and R. Abitbol, "Beyond Black-Box Benchmarking: Observability, Analytics, and Optimization of Agentic Systems," arXiv:2503.06745v1, 2025.

[32] M. Zhuge, C. Zhao, D. R. Ashley, W. Wang, D. Khizbullin, Y. Xiong, Z. Liu, E. Chang, R. Krishnamoorthi, Y. Tian, Y. Shi, V. Chandra, and J. Schmidhuber, "Agent-as-a-Judge: Evaluate Agents with Agents," arXiv:2410.10934v2, 2024.

[33] S. Akshathala, B. Adnan, M. Ramesh, K. Vaidhyanathan, B. Muhammed, and K. Parthasarathy, "Beyond Task Completion: An Assessment Framework for Evaluating Agentic AI Systems," *International Workshop on Agentic Engineering (AGENT '26)*, Rio de Janeiro, Brazil, 2026. doi: 10.1145/3786167.3788414.

[34] M. Gritta, D. Paul, X. Li, L. Shang, J. Wang, and G. Lampouras, "Process Evaluation for Agentic Systems," *Findings of the Association for Computational Linguistics: EACL 2026*, pp. 2678-2692, 2026.

[35] M. Baduwal and P. Paudel, "Evaluating Agentic Artificial Intelligence: A Comprehensive Survey of Metrics, Benchmarks, and Methodologies," *TechRxiv*, 27 February 2026. doi: 10.36227/techrxiv.177162480.04513202/v2.

[36] [CrewAI documentation: open-source framework for role-based multi-agent crews and flows](#)

[37] [Microsoft Agent Framework: open-source successor to AutoGen and Semantic Kernel](#)

[38] [Google Agent Development Kit \(ADK\): open-source, model-agnostic agent framework](#)

[39] [Anthropic Claude Agent SDK overview: production agent primitives, tools, MCP, and hooks](#)

Appendix A. Hackathon scope and judging rubric (Advanced Track final, 12 June 2026)

This appendix turns the product vision above into a focused, build-in-a-day challenge and the rubric the panel will use. It is an addition; nothing in Sections 1 to 12 is changed.

A.1 Hackathon - Day-one scope (what you build in the final)

- Build a working agent for two scenarios: a **suspicious refund request** (the fraud-pause path) and a **duplicate or unrecognized charge**.
- Use mock data for orders, payments, shipments, and prior cases. Do not connect live financial systems.
- Implement the full loop: verify identity (mocked), retrieve evidence, screen for fraud risk, apply a versioned policy, propose a remedy, pause and route to a human-approval gate on risk, then write an audit log.
- Use three to five least-privilege tools with read, prepare, and execute kept separate (for example `get_order`, `get_payments`, `get_shipment`, `create_risk_case`, `prepare_refund`).
- Deliver a five-minute live demo of both journeys, the visible audit trail, and a short pitch that is honest about what is real versus mocked.
- Out of scope for the day: live payment integration, full compliance, a production evaluation harness, voice, and multi-channel intake.

A.2 Judging rubric (100 points)

Each criterion is scored within its band. Two or three judges score independently and the average is the team score. Ties are broken first by Working agent loop, then by Fraud detection.

Criterion (weight)	Excellent (full marks)	Good (partial)	Needs work (low)
Working agent loop (25)	Autonomous multi-step loop: plans, calls tools in sequence, holds state across turns, and recovers from a failed or empty result.	Multi-step but brittle; some steps hard-coded; limited state.	Single-turn or scripted; no real planning or state.
Fraud detection + human-in-the-loop (20)	Detects risk signals, pauses the autonomous action, routes to a human-approval gate, and never makes unsupported accusations to the customer.	Detects some risk and flags it, but the pause or approval is weak or fully manual.	No risk screening, or proceeds with a sensitive action unchecked.
Tool-call correctness + least-privilege (15)	Correct tools and parameters; read, prepare, and execute are separated; amount, policy, and idempotency checks run before any prepare-refund.	Mostly correct; partial separation; some checks present.	Wrong or over-privileged tools; executes money actions without checks.
Auditability (15)	Full trajectory log shown in the demo: tool calls, evidence, risk score, decision rule, approval, and customer notice.	Partial log; missing parts of the chain.	Little or no audit trail.
Policy grounding + remedy correctness (10)	Remedy follows a versioned policy, is correct for the facts, and eligibility is validated.	Reasonable remedy; policy loosely applied.	Arbitrary or incorrect remedy.
Demo clarity + honesty (10)	Clear five-minute demo of both journeys; explicit about what is real versus mocked; crisp narrative.	Demo works but is unclear or overclaims.	Confusing or non-working demo.
Innovation / stretch (5)	A real extra: a second domain, prompt-injection defense, a trace or eval metric, or a novel UX.	A minor extra.	None.

Total: 100. Suggested cut: a credible finalist scores 60 or above with a working agent loop and a clear fraud-pause.